

Extracting relations between diseases, treatments and symptoms from unstructured biomedical text

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ABSTRACT

Novel articles containing modern advancements and discoveries in the medical industry are being published online daily. This study aims at creating a biomedical relation extraction tool, specialized in identifying disease-symptom and disease-treatment relations from unstructured text in the medical domain. The system should be able to automatically infer and model structured knowledge from newly published medical sources, specifically focused on diabetes and heart disease as use cases due to their predominant presence in the local sphere. However, the solution will be scalable to eventually cater for other diseases.

1. INTRODUCTION

Medical knowledge can often be found in narrative form such as medical textbooks, clinical records or online published medical reports. Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) which can be used to extract meaningful information from textual data. Numerous medical reports are being published online daily on websites such as PubMed, containing information on advancements in the medical industry. Automated information extraction tools [1, 2, 3] could be used to infer new structured knowledge from such sources that could be useful for researchers or medical students. It should be noted that information extraction tools rely heavily on annotated corpora, which are commonly scarce and contain sensitive data, particularly electronic medical records (EMR) which are digital hospital records containing valuable information on patients such as their medical history and assigned treatments [4]. There have been certain competitions [5] that released annotated corpora for participants to use for tackling NLP tasks but due to privacy issues and labour intensive work, there has been a small amount of corpora available. As a result, there have been few advancements of such tools [1, 2, 3] in the medical industry which are openly available and accurate.

Named entity recognition (NER) is a sub-task of information extraction that focuses on extracting concepts from unstructured text such as names or locations. Biomedical named-entity recognition (BM-NER) focuses on the recognition of medical entities such as genes, proteins [6, 7] or disease names [5]. Early-based BM-NER systems are rule-based [1, 2, 3] and rely on medical vocabularies such as the Unified Medical Language System (UMLS) [8]. Generally, more recent tools employ machine learning techniques [9, 10, 11], specifically conditional random fields (CRF) which are discriminative classifiers that use contextual information

from neighbouring labels to predict sequences.

Relation extraction (RE) is a task that focuses on extracting semantic relations between two or more entities. Relation extraction systems can reduce the time spent by experts on labour-intensive tasks such as the conversion of information from unstructured sources into structured form. These systems are typically pattern-based [12] or machine learning based [13, 14]. Pattern-based approaches do not require annotated data and use hand-coded rules which require the involvement of domain experts. Machine learning based approaches require annotated data and model relation extraction as a classification problem. Popular machine learning approaches use support vector machines (SVM) which are models that plot features in a multidimensional space and differentiates their class labels through the construction of hyper-planes.

Diabetes and heart diseases are one of the most common health issues in Malta. According to a news article¹, “one in eight of Maltese adults aged 25 to 64 years suffer from diabetes, of whom approximately 10,000 are unaware of the disease”. In addition to this, another article² mentions that in 2015, over 38 percent of deaths were linked to diseases related to the cardiovascular system. Both articles mention that the population are not aware of the risks from these diseases, highlighting the need of education tools to demonstrate newer symptoms that could occur as well as novel treatments that have been introduced.

This research will evolve around investigating approaches that allow for the extraction of entities and relations from biomedical literature, more specifically focused on two diseases, namely diabetes and cardiovascular diseases. The system should be able to automatically infer and model structured knowledge from newly published medical sources on diabetes and heart disease.

2. AIMS AND OBJECTIVES

This study aims at identifying medical entities and extracting disease-symptom and disease-treatment relations from unstructured biomedical text. This goal will be addressed by seeking solutions to the following two challenges;

- Challenge 1: BM-NER is a necessary challenge to address in order to build a successful information extraction tool. This challenge will be tackled by creating a

¹<https://timesofmalta.com/articles/view/One-in-eight-Maltese-adults-suffering-from-diabetes.663045>

²<https://timesofmalta.com/articles/view/heart-disease-know-your-risks.691576>

machine learning model that is able to locate and distinguish between disease, treatment and symptom entities within given unstructured data. Since the dataset will provide annotations of problems, test and treatments, the system will need to be able to differentiate whether an identified problem is a disease or a symptom.

- Challenge 2: Biomedical relation extraction is another challenge to focus on for the construction of the finalized tool. This challenge will be addressed by analysing different methodologies proposed in the previous literature's and combining their optimal approaches to identify the appropriate relations. The disease-symptom relation aims at establishing whether symptom causes a particular disease. The disease-treatment relation determines whether a treatment cures a certain disease.

3. LITERATURE REVIEW

In 2010, the i2b2/VA Workshop [5] presented an NLP challenge consisting of three tasks; concept extraction, assertion classification, and relation extraction of medical entities using an annotated corpus of de-identified clinical reports. The following section will discuss the best performing systems of the first and last task of the challenge and compare them to other developed systems for BM-NER and RE.

3.1 Biomedical Named Entity Recognition

Early based clinical NER systems used rule based-methods [1, 2, 3] and relied on existing medical vocabularies such as the UMLS [8]. Although rule-based NER systems produce effective results, they remain domain specific and require extensive manual effort. As a result, most BM-NER systems employ machine learning (ML) approaches since they do not require human effort and can be reprogrammed to handle other domains [4]. However, ML models require large amounts of annotated corpora in order to produce beneficial results.

During the concept extraction task of the i2b2 2010 challenge [5], participants were tasked with the extraction of three medical entities from the provided dataset. The three entities to be extracted were related to medical problems, treatments and tests. A majority of the participants of the challenge that achieved the best results [9, 10, 11] tackled the task using conditional random fields (CRF). An overview of the top solutions and results presented by the participants will be described in the following section.

DeBruijn et al. [15] was one of the exceptions to the CRF-based solutions and created a concept extraction system that generated the highest F-measure result in the challenge for the first task (0.852). The system was built using Semi Markov Models (SMM) instead of the typical BIO (Begin, Inside, Outside scheme) tagging formalism and trained using an online learning algorithm referred to as the Passive Aggressive (PA) algorithm. A SMM was used due to its ability to tag multi-token spans of text and allows the system to have a larger flexibility during feature construction. The PA algorithm was used instead of a CRF-based approach to enhance the discovery task and to improve memory performance especially concerning large feature sets. Additionally, [15] highlighted that the use of external sources such as UMLS for semantic and syntactic tagging respectively,

improved the F-measure of their system.

Jiang et al. [9] constructed a hybrid concept extraction system called MedNet by combining a heuristic rule based approach with a CRF-based NER module. They also constructed an assertion classifier for the second task of the challenge [5] using the same concept but incorporating an SVM-based NER module. Through experimentation, [9] discovered that the CRF system outperformed the SVM system. Similarly to DeBruijn et al. [15], Jiang M. et al used various feature types, syntactic and semantic information from external sources such as MedLEE [3], KnowledgeMap [2] and UMLS.

Kang et al. [10] tackled the concept extraction challenge by tagging the provided data using a total of six different NER tools and chunkers. All six tools were trained on the dataset and the annotated results of each system was passed through a simple voting scheme to obtain a combined annotated result. Three of the NER tools that were used (ABNER [6], OpenNLPner³, StanfordNer⁴) produced the highest F-measure results individually (0.741, 0.776, 0.768), however the combination of all six tools allowed the system to achieve a higher F-measure score of 0.821.

Gurulingappa et al. [5] approached the challenge by enhancing textual features using a rule-based NER, then feeding its output to train a CRF model. The system obtained an F-measure of 0.818. Torri and Liu [11] explored the use of a tool called BioTagger-GM [7], which was previously used for the recognition of gene/protein names. The tool enhances features using dictionary lookup methods from terminology resources such as UMLS. Torri and Liu conclude that the performance of the system is improved when using concept dictionaries. Furthermore, [11] emphasize the importance of a large quality corpora and dictionary resources for the development of ML models for such tasks.

To review, the participants of the challenge [5] demonstrated that systems achieved higher results using rich textual features and the inclusion of knowledge rich sources such as the UMLS. The biggest obstacles of the task were discovered to be the labelling of abbreviations and highly descriptive concept phrases.

To address the lack of annotated corpora for medical NER systems, Zhang et al. [16] present an unsupervised solution to BM-NER, called cubNER. Their system explores a signature-based approach where signature vector representations are generated for target entity classes (such as Treatment). The UMLS is used to extract seed terms for each class. Candidate named entities are extracted from a given text through the use of NP-chunkers and Inverse Document Frequency (IDF), and then classified using cosine similarity between the generated signature of the entity and a signature of a target entity class. Ghiasvand et al. [17] propose an improved version of cubNER [16] whereby they construct a system that incorporates the same signature-based approach but employs machine learning models that perform self-iterations and are trained using annotations obtained from the raw text corpus and UMLS. Their improved system is able to recognize entities that occur in varying contexts, employ full parsing and consider entire noun phrases. As exemplified by both papers, although unsupervised methods address the problem of insufficient data in the clinical do-

³<https://opennlp.apache.org/>

⁴<https://nlp.stanford.edu/software/CRF-NER.shtml>

main, both systems generated low F-measure results (0.241 and 0.331) compared to the machine learning approaches discovered above, highlighting the importance of an annotated dataset.

Wu, Yonghui et al. examined the impact of word embeddings and deduced that they can be used as features to improve the performance of ML models for NER [18, 19]. As a result of this discovery, they developed a CNN based NER method that uses word embeddings and applied it to Chinese Clinical notes [18]. Their system generated extremely high results with a F1-score of 0.928. Subsequently, they compared the performance of two popular deep learning models, a CNN and RNN to extract clinical concepts from English clinical text [19] and specifically using the i2b2 challenge dataset [5]. They compare the deep learning models to three baseline CRF systems with additional features at each level and two state-of-the-art systems from the i2b2 challenge, one of which is the system that generated the highest result [15]. Wu, Yonghui et al. [19] used the MIMIC II dataset⁵ for the discovery of word embeddings and the i2b2 2010 dataset for training and testing. They discovered that the RNN was able to outperform all other systems, achieving a high F-measure score of 0.828. The deep learning models were able to capture long-term dependencies and learn high-level features automatically without the need of human feature engineering.

Another deep learning approach to BM-NER was explored by Liu, Zengjian et al. [20] who investigated the performance of a Long Short-Term Memory (LSTM), a variant of the RNN. In this research, the datasets from the i2b2 2010 challenge as well as the 2012 and 2014 i2b2 challenges [21, 22, 23] to train and evaluate their model. Liu, Zengjian et al. deduce that their system is able to outperform CRF models and discover that the use of character-level word representations into the input layer of LSTM improved results.

In Table 1, a summary of the research considered for BM-NER is presented. As can be seen from the results, the systems that used deep learning techniques outperformed the other systems. On the other hand, unsupervised systems generated the lowest score, outlining the importance of having annotated corpora for BM-NER tasks.

3.2 Biomedical Relation Extraction

In the third task of the i2b2 2010 challenge [5], participants were asked to identify relations between the three entity types that were recognized in the concept extraction task. The relations could be categorized into three main categories; Problem-Treatment, Problem-Test and Problem-Problem relations. A Problem-Treatment is a category of relations that indicate that some treatment improves some medical problem (TriP), for instance the problem of hypertension was controlled using hydrochlorothiazide. In the case of the Problem-Test category, this includes relations that indicate that some test is able to reveal some medical problem (TeRP). A typical relation example in this case is that of a chest X-Ray (test) that reveals a broken ribcage. On the other hand, the Problem-Problem category includes relations that associate a medical problem with some other medical problem (PIP), for instance, Azotemia is presumed to be a secondary cause of sepsis (blood infection). A detailed table of these concept relations can be found in [5]. It must be noted that the task also includes the extractions

of pairs of entities in the same sentence, even if they do not match any of the specified sub-concept relations. The following section will proceed by discussing the methods and results obtained by the top performing proposed systems in the challenge.

Rink et al. approached the task by implementing an SVM to identify and assign semantic types to relations between the concepts [24]. Their system obtained the highest F-measure score of the relation extraction challenge [5] (0.737). Their approach highlights the importance of using rich sets of lexical and context features to strengthen ML models. They experimented on their relation extraction system by creating one which is provided with manually annotated concept and assertion annotations and one where the annotations are not provided. Based on their experiments, [24] concluded that automatically annotating concepts and assertions degraded the systems performance, highlighting the impact of the quality of concept discovery in relation extraction. They also deduced that the use of external knowledge sources, such as UMLS, and incorporation of joint learning could improve results. It must be noted that each feature used in the SVM benefited the relation extraction results but certain features produced more useful information to the extraction of specific relations. For instance, exploiting Wikipedia's hyperlink structure and category hierarchy benefited the PIP relation by enhancing information on whether two concepts are strongly associated.

DeBruijn et al. proposed a system [15] that achieved the second highest F1-score (0.731) for the relation extraction task [5]. The system uses maximum entropy (ME) as a classification framework and classifies each relation independently, similarly to [24]. The utilization of parsing trees and balancing of positive/negative ratios for training set samples were found to be amongst the most beneficial features that were used. DeBruijn et al. developed three system variations for the relation extraction task; the first variant uses augmented-baseline features and dependency features. The second variant uses semantic features from MEDLINE to calculate the Pointwise Mutual Information (PMI) between two concepts to approximate the extent to which they are related, while the last variant generated the highest result (0.731) and employs semi-supervised learning by applying bootstrapping on unlabeled data using their system [15] that was previously developed for the first task of i2b2 challenge on concept extraction.

Grouin et al. [25] developed a hybrid system that combines an SVM with manually constructed linguistic patterns. The linguistic patterns enhance results due to the fact that certain relation types do have sufficient annotations to train the classifier and adds confidence to the results by confirming automatically-induced relations. The research explores three variants of their system; the first variant constructs patterns to identify four relations which had few examples in the training data (TeCP, TriP, TrNAP, and TrWP). The second variant employs a supervised approach and trains the SVM from text that was syntactically simplified using certain techniques such as the removal of syntactic phrases between candidate concepts. The last variant is a combination of the first two and produced the highest result (0.709). Grouin et al. concluded that their system obtain higher results when extracting positive relations when compared to the extraction of negation relations and performed better with more training examples.

⁵<https://mimic.physionet.org/>

Table 1: Overview of BM-NER systems

Authors	Type	Method	Dataset	F-Measure
Wu, Yongui et al. [18]	Supervised	CNN	Chinese Clinical Records	0.928
Wu, Yongui et al. [19]	Supervised	RNN and CNN	I2b2 2010 and MIMIC II	0.859 and 0.823
Liu, Zengjian et al. [16]	Supervised	LSTM	I2b2 2010, 2012 and 2014	0.858, 0.923 and 0.944
DeBruijn et al. [15]	Semi-supervised	SMM and PA	I2b2 2010	0.852
Jian et al. [9]	Hybrid	CRF	I2b2 2010	0.839
Kang et al. [10]	Hybrid	CRF, HMM	I2b2 2010	0.821
Gurulingappa et al. [5]	Supervised	CRF	I2b2 2010	0.818
Torri and Liu [11]	Supervised	CRF	I2b2 2010	0.813
Ghiasvand et al. [17]	Unsupervised	Signature-based	I2b2 2010 and SemEval 2014	0.331 and 0.693
Zhang et al. [16]	Unsupervised	Signature-based	I2b2 2010	0.241

Patrick et al. developed a pipeline information extraction system [26] that combines solutions they presented for each task of the challenge and employs a combination of ML and rule-based approaches. Their system uses clinical resources such as UMLS and SNOMED-CT⁶ as well as a general English lexicon, that was used for lexical verification and spelling correction. SNOMED-CT refers to the systematic nomenclature of medicine—clinical terms and contains a collection of medical terms and general terminology for EHR’s, including coverage on diagnoses, clinical findings and symptoms. For the relation extraction task, the system [26] uses an SVM and various local context and semantic features. Results indicate that the TeRP, TrAP and PIP relations had the highest F-scores. Patrick et al. conclude that the small dataset created a large drop in the F-score of the system due to unseen examples. On the other hand, their model generated high results while maintaining simplicity without incorporating large features spaces like other participants.

Overall, the third task of the i2b2 challenge illustrated that most effective relation extraction systems adopt the SVM approach. It was discovered that the provided corpus had several concept pairs that had no relations, hence, certain systems chose to remove those relations prior to identifying relations [15]. It should be noted that certain systems benefited from down-sampling and the augmentation of hand-built rules [25]. Most importantly, the papers highlighted that the performance of their system could have significantly improved if provided with more labelled examples, indicating the need of domain knowledge in the annotation of these examples.

Frunza et al. focus on the identification of relations between diseases and treatments [27] using the BioTextc dataset [28], which contains a collection of 1,000 manually annotated and randomly selected MEDLINE abstracts and titles. The paper focuses on the extraction of three relations; Cure, Prevent and Side Effect. The paper uses a total of six classifiers and predicts the majority class in the training data. The system also incorporates a combination of representation techniques, specifically the bag-of-words (BOW) representation, UMLS representation and NLP and biomedical concepts representation. Frunza et al. experiment on the identification of the three relations with three different settings and deduce that the final setting which uses a single model trained on a dataset containing sentences, annotates with three relations to obtain the best results. Keerthega et al. [29] explore a similar approach utilising the same dataset [28] and survey the use of SVM, RBF Networks and Sequential Minimal Optimizer (SMO) as classifiers. Experiments

indicate that the SVM generated the highest accuracy. On the other hand, the SMO achieved a higher F-measure of 0.860.

In addition to their previous work, Frunza et al. developed a supervised SVM-based classifier [13] and experiment with more informative representations to influence the results of the ML algorithm on the i2b2 2010 dataset [5]. The developed system produces an F-measure score higher than the average score of the participants of the challenge (0.862). In future work, [13] plan to add features and additional information that is specific for each concept and relation.

Lv, Xinbo, et al. developed a supervised CRF-based machine learning model which predicts tags of relationships based on context features [14]. The paper implements a DL model for feature learning and the discovery of word embeddings in the context of concept relations due to the importance of features in ML modelling. Additionally, [14] adopt DL models to learn and optimize word features instead of using a BOW representation. Through experimentation, Lv, Xinbo, et al. evaluated three methods and conclude that the additional use of autoencoders and sparsity limitation generated the highest result (0.820) and outperforms the best systems in the i2b2 2010 relation extraction task challenge [5]. As future improvement, the authors propose exploring the use of knowledge bases such as UMLS to detect relations between hypernyms and implement them into the model.

Hassan, Mohsen, et al. [30] propose a novel method known as SPARE (Syntactic Pattern for Relationship Extraction) for automatic extraction of disease-symptom relationships, particularly focused on rare diseases. SPARE learns patterns from dependency graphs of given text which have at least one symptom and disease. During generation, patterns undergo a generalization process which affects precision and recall. The system then selects patterns based on their quality and specificity which are then used to extract relations. In conjunction with the fact that the system is able to discover new relationships, it should be noted that patterns with a high level of specificity provide the system with the potential to discover new and complex symptoms.

In conclusion, existing research has highlighted the limitation in performance due to the availability of a small corpus. In Table 2, an overview of the results obtained by the papers discussed in the section is presented. Overall, the majority of the top performing systems employed machine learning supervised approaches and the use of SVM was the most popular method and produced some of the highest scores. Certain papers outlined the importance of the use of rich features for the training of the ML models [24, 15]. A few papers also mentioned that the exploration of implementing knowledge bases such as UMLS could generate higher results

⁶<http://www.snomed.org/>

for the system [14].

4. PROPOSED SOLUTION

The proposed system will consist of two main components; the BM-NER system and the biomedical relation extraction system. Access to the i2b2 2010 dataset [5] and UMLS [8] has been requested and approved and will be used in the creation of the system. The planned technique for each component will be described below.

4.1 Biomedical Named Entity Recognition

A BM-NER system will be implemented by constructing a CRF model by exploring various context features similar to papers discussed previously [15, 9] and the UMLS for additional semantic information. Through initial research, it was found that CRF-based solutions tend to outperform other solutions, hence the same technique will be adopted as a starting point. Access to the UMLS was granted by registering and requesting a license from the U.S. National Library of Medicine under certain agreements⁷. The system will be trained on data from the 2010 and 2012 i2b2 datasets [5, 21] which were obtained by requesting online access from Harvard Medical School given certain data use agreements. Since the i2b2 datasets only labels problems, treatments and tests, problem labels could consist of symptoms and disease names. To solve this issue, [31] describes a method on exploiting the UMLS for categorizing concepts representing signs and symptoms using UMLS semantic types.

4.2 Biomedical Relation Extraction

A biomedical relation extraction tool will be constructed by creating an SVM model which produced some of the highest results in the research [13, 24, 25, 26]. An SVM model was chosen as it was discovered to be the most popular approach that yields high results in comparison to other systems. The system will be focused on identifying TrIP (treatment improves medical problem), TrAP (treatment is administered for medical problem) and PIP (medical problem indicates medical problem) relations for the 2010 i2b2 dataset. Similarly to [14, 19], the paper will explore the use of word embeddings and evaluate their performance when used as input data for the SVM. Since most systems in the challenge mentioned that a lack of corpora affected their results, the paper will explore the integration of unsupervised cause-effect relation extraction systems [32] since the PIP relation can be modelled as a causal relation.

5. EVALUATION PLAN

The system will be evaluated using precision, recall and F-measure. Previous works evaluated their systems using annotated clinical records [18] or manually annotated MEDLINE abstracts [27, 29] and split the annotated data into training and testing sets for evaluation. Due to privacy issues and the need of medical experts, annotated medical related data is difficult to obtain. As a result, the i2b2 2010 challenge [5] will be used as a gold standard for the systems performance evaluation for the BM-NER and biomedical relation extraction module, since access to the training and testing datasets has been obtained and novel research [19, 14] still use it to evaluate their systems.

⁷<https://uts.nlm.nih.gov/license.html>

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Table 2: Overview of Biomedical Relation Extraction systems

Authors	Type	Method	Dataset	F-Measure
Frunza et al. [13]	Supervised	SVM	I2b2 2010	0.862
Lv, Xinbo, et al. [14]	Supervised	CRF and deep learning	I2b2 2010	0.82
Rink et al. [24]	Supervised	SVM	I2b2 2010	0.737
DeBrujin et al. [15]	Semi-supervised	ME	I2b2 2010	0.731
Grouin et al. [25]	Hybrid	SVM and linguistic patterns	I2b2 2010	0.709
Patrick et al. [26]	Supervised	SVM	I2b2 2010	0.702
Frunza et al. [27]	Supervised	Combination of classifiers	BioTextc	Cure, Prevent and Side Effect achieves an F-measure of 0.986, 1.000 , 0.889 respectively
Keerrthegea et al. [29]	Supervised	SMO	BioTextc	0.86
Hassan et al. [30]	Supervised	Syntactic patterns and DG's	PubMed abstracts	0.557

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